

Space Resource Economic Analysis Toolkit: The Case for Commercial Lunar Ice Mining

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1.0 EXECUTIVE SUMMARY

An integrated engineering and financial modeling approach and Excel toolkit has been developed and used to evaluate the potential for private sector investment in space resource development, and to assess possible roles of the public sector in fostering private interest. This report presents the modeling approach and its results for a transportation service using propellant extracted from lunar regolith to provide transfer between low Earth orbit (LEO) and geosynchronous orbit (GEO).

The modeling approach started with the definition of an economic case study, including a thorough analysis of the customer base leading to the development of a demand model. These inputs form the foundation for developing an engineering model of a modular, scalable commercial space architecture designed to meet demand. A cost model derived non-recurring, recurring and operations costs, which became inputs for a 'standard' financial model, as used in any commercial business plan. This financial model generated pro forma financial statements, calculated the amount of capitalization required, and generated return on equity calculations using two valuation metrics of direct interest to private investors: market enterprise value and multiples of key financial measures. Finally, sensitivity analysis with respect to key strategic, market and technological inputs helped to further explore the conditions for financial viability.

This modeling approach is illustrated on a lunar propellant case study. Two separate architectures were developed that model the conversion of water held in permanently shadowed lunar craters into propellant for use in near-Earth space transportation, in particular to convey payloads from low Earth orbit (LEO) to geosynchronous Earth orbit (GEO). Both models generated nearly identical economic results, identifying the technical and financial conditions under which the architectures could become commercially attractive.

Production and transportation system masses were estimated for each of the two architectures, and cost analysis was made using the NAFCOM and SOCM cost models. Data from the cost models were analyzed using standard financial analysis tools to determine under what conditions the architectures might become commercially viable. Analysis of the architectural assumptions were used to identify the principal areas for further research, which include technological development of lunar mining and water extraction systems, power systems, reusable space transportation systems, and orbital propellant depots. The architectures and their commercial viability are strongly sensitive to the assumed concentration of ice in the lunar deposits, suggesting that further lunar exploration to determine whether higher-grade deposits exist could be economically justified. Business assumptions, in particular the implications of government support of the R&D required for system development, were also explored.

This use of the modeling approach on two architectural variants of a lunar propellant case study demonstrates how to rapidly test various assumptions and identify interesting architectural options, key areas for investment in exploration and technology, or innovative business approaches that could produce an economically viable industry. The same approach could be used to evaluate other possible commercial ventures in space, providing feedback about the respective roles of NASA and the private sector in space resource development and solar system exploration.

2.0 INTRODUCTION

NASA is studying options for expanded solar system exploration, and the NASA Exploration Team (NExT) is exploring alternative mission architectures and enabling technologies. An important consideration in these studies is the potential role for the private sector in supporting solar system exploration, and how NASA can leverage private sector capabilities to achieve its objectives more cost-efficiently. However, while there is a broad consensus that private sector participation is desirable, there has been a limited amount of work within NASA to address this question from the perspective of the private sector. Chartered by NExT, the Jet Propulsion Laboratory (JPL) chose the Colorado School of Mines (CSM), and CSP Associates, Inc. to help them develop an economic modeling tool to complement engineering studies by simulating the private sector investor's point of view.

Although qualitative arguments can be made for the benefits of on-orbit servicing, space manufacturing, planetary surface mining, etc, no realistic conclusion can be reached without quantitative analysis of the financial viability of a private venture. In order to reach solid conclusions regarding economic feasibility a flexible, integrated financial and engineering model was required. The multi-disciplinary science, engineering and financial team was gathered in order to model all aspects of the proposed commercial venture, and bridge the gap between NASA and the private sector. The model developed and described herein was applied to a specific case study of commercial lunar propellant utilization. However, it is believed that this approach and especially the modeling approach and toolkit will be useful for many other architectures and space-based ventures.

A scenario to sell in-space transport based on lunar propellant was proposed as the first case study to examine the potential for space resource economic viability. Smitherman (2001) showed that there is a significant market for LEO-to-GEO transport based on cryogenic H_2/O_2 propellants. Although that study assumed Earth-based propellant, the Moon is actually much closer to LEO in terms of delta-V requirements than the Earth's surface. In addition, the Lunar Prospector's mission data indicated sufficient concentration of hydrogen (presumed to be in the form of water ice) to form the basis for lunar in-situ mining activities to provide a source of H_2/O_2 propellants. Such propellant could also be very useful to NASA's solar system exploration missions if provided at the Earth-Moon L1 Lagrange point, highlighting the potential for public as well as private interest. Finally, preliminary engineering analysis based on known terrestrial mining and processing technologies showed that the required architecture mass would be much smaller than the total mass of propellant it could produce and deliver to L1 or LEO. Based on these preliminary checks, the team set out to analyze LEO-to-GEO transport using lunar-based propellants.

Note that the presence of ice on the Moon, its concentration and abundance, and its physical properties are conjectures at this point. The technology for working within permanently shadowed craters on the lunar surface, where temperatures are less than 100 K, is at a conceptual state of development at best. Also, there is no present customer that can readily accept propellant that could be produced from these water deposits, although propellants are used in substantial quantities to convey payloads such as communications satellites from LEO to GEO. The model's assumptions are described, but they should be

taken as propositions that remain to be demonstrated, not facts. Nevertheless, analysis of the architecture model allows one to discuss which of the assumptions are most critical and to provide some guidance for further exploration and technology development.

2.1 THE BASIS FOR SPACE RESOURCE VALUE

A number of studies have shown the potential offered by space resource utilization for space missions. Eagle Engineering (EEI, 1988) conducted a systematic study of the potential for using lunar oxygen in support of lunar missions. Other studies have described similar applications for Mars missions (e.g. NASA, 2001). Duke (1998) analyzed possible lunar ice extraction techniques and (Rice, 2000) showed how using this ice to produce lunar-based cryogenic H_2/O_2 propellants would reduce the Earth launch mass for a reference lunar outpost mission by up to 68%. Based on similar assumptions of NASA lunar transportation requirements, Nelson (2001) calculated the price a private venture would need to charge for transfer of cargo and astronauts to the Moon. Borowski (1997) studied the improvements in lunar transportation that could be brought about by nuclear thermal propulsion. For low Earth launch costs and given transportation requirements, Stancati (1999) showed that using lunar-based LOX and LH_2 and nuclear thermal propulsion could enable technical improvements in Earth launch mass of up to 51%, but with negligible cost improvements. These are only a few examples of a wealth of interesting engineering studies that characterize what we might call the “potential for space resources supply”.

Although much less numerous, there also have been a few studies to characterize the “potential for space resources demand”. The commercial space transportation study (CSTS, 1994) carried out a systematic, quantified analysis of potential markets for future launch services. Smitherman (2001) quantified the demand for cryogenic propellants in LEO for LEO-to-GEO transfer. Between these two bodies of research and analysis (the “supply” and the “demand”), there is a clear gap: Among all the architectures proposed for space resources development, do any suggest (financially) viable private ventures?

High-level definition of the lunar propellant case study began with a combination of engineering and financial “common sense.” First, an identifiable, predictable market must exist. For example, the projected market for in-space transportation services was derived from current government and commercial launch demand to various orbital destinations. Second, there must be good potential for market capture, i.e. a potential for providing the resource cheaper than direct or functionally equivalent competitors. In the case of LEO-to-GEO transfer based on lunar propellant, two already-established competitors exist that guide initial pricing assumptions: (1) direct launch into GEO, and (2) use of Earth-based propellants transported to a LEO fuel depot (e.g., Smitherman, 2001).

Because a commercially viability venture relies on private investment, a model that represents costs and benefits in private sector investors’ terms was needed. To do this, an engineering system architecture must be developed, costs of development, production and operation of the system must be estimated, and a reasonable set of market assumptions adopted. The integrated model then can be used to determine financial feasibility.

2.2 TRANSPORTATION AND LOGISTICS

Transportation in space is a major consideration because of its high cost. Indeed, high launch costs are one of the primary reasons an in-space fuel source has value. While the unit cost of in-space production of a resource can be expected to be much higher than on Earth, its in-space transportation cost from the place of production to the place of use in space has the potential for being much lower than launch cost from Earth. A secondary argument for value is the potential for reuse and refueling of orbital transfer vehicles, which can be compared with the current practice of expending launch vehicle elements after their first use. If inexpensive propellant can be provided in space, these otherwise disposable vehicles may gain in value.

Non-engineering professionals often think of space transportation in terms of distance. The idea that LEO is closer to Earth than the Moon is only true in terms of distance (LEO lies roughly 0.1% of the distance to the Moon - see Table 2.1). A more relevant variable, and the one most commonly reported in the aerospace literature, is the change in velocity required to reach a specific orbit (ΔV , typically reported in km/s). However, the best metric for the energy it takes to get from one orbit to another, and therefore for the amount of propellant needed, can be found by squaring ΔV (ΔV^2 is reported in units of mega joules per kilogram – a direct measurement of energy). Table 2.1 shows distance, ΔV and ΔV^2 for the Earth-Moon system. Note that by using the ΔV^2 metric, LEO is 83% of the way to the Moon. Add the efficiency of aerobraking, and LEO is over 96% of the way to the Moon (the ΔV to aerobrake from the Earth-Moon L1 Lagrangian point to LEO is only 500 m/s, compared with 4.6 km/sec for a propulsive maneuver). This clearly demonstrates the transportation energy advantage that the Moon holds over the Earth (see Figures 2.1 - 2.3 for a graphical sketch of the Earth-Moon system in ΔV^2 scale), for operations in LEO or higher orbits. Values in Table 2.1 assume the use of Hohmann transfers, which are typical of high-thrust systems (advantages of high-thrust cryogenic systems over low thrust ion propulsion include faster transit times, technological heritage and lower costs).

Table 2.1. Comparison of scales in the Earth-Moon system.

Location	Distance (km)		Delta V (km/sec/kg)		Delta V ² (MJ/kg)	
	increment	cumulative	increment	cumulative	increment	cumulative
Earth-LEO	400	400	9.5	9.5	90.3	90.3
LEO-GEO	29022	29422	3.8	13.3	14.4	104.7
GEO-L1	256100	285522	0.8	14.1	0.6	105.3
L1-LLO	92400	377922	0.9	15.0	0.8	106.1
LLO-Moon	100	378022	1.6	16.6	2.6	108.7

Note that the values for ΔV have been calculated for most known sources of space resource materials (exceptions include unidentified asteroids). Transportation systems in space must carry their own propellants and it is straightforward to take a design for a transfer vehicle, determine its performance, and utilize the rocket equation to determine the amounts of propellant needed to make a particular transfer.

Figure 2.1. Earth-Moon Transportation Energy using ΔV scale ($1'' = 4.3 \text{ km/s/kg}$).

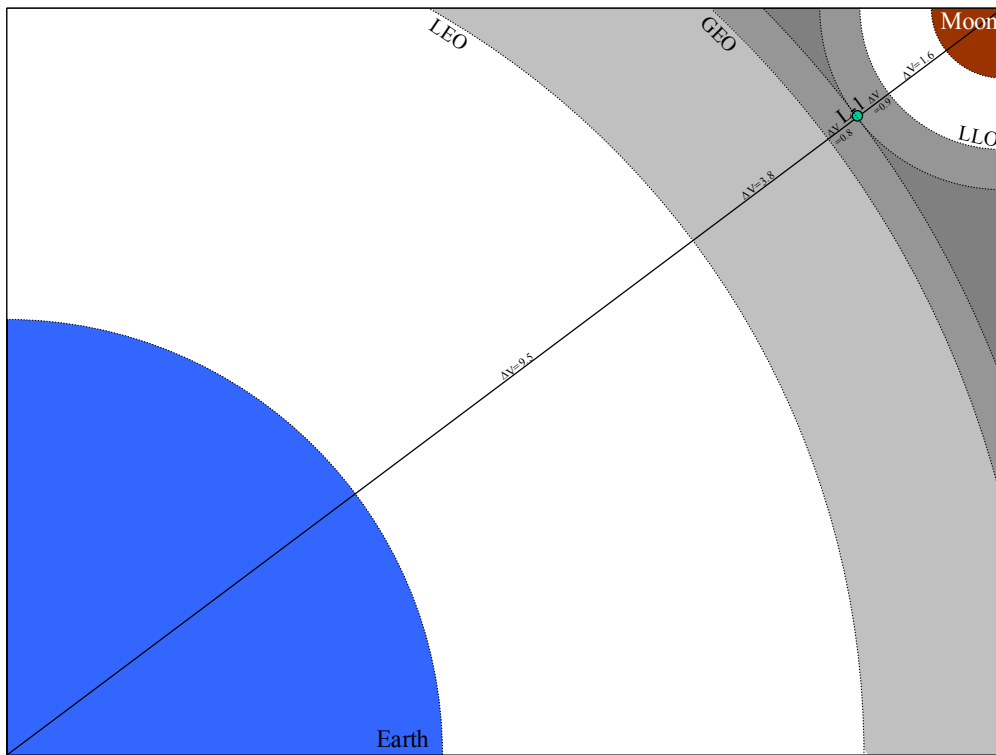


Figure 2.2. Earth-Moon Transportation Energy using ΔV^2 scale ($1'' = 32 \text{ MJ/Kg}$).

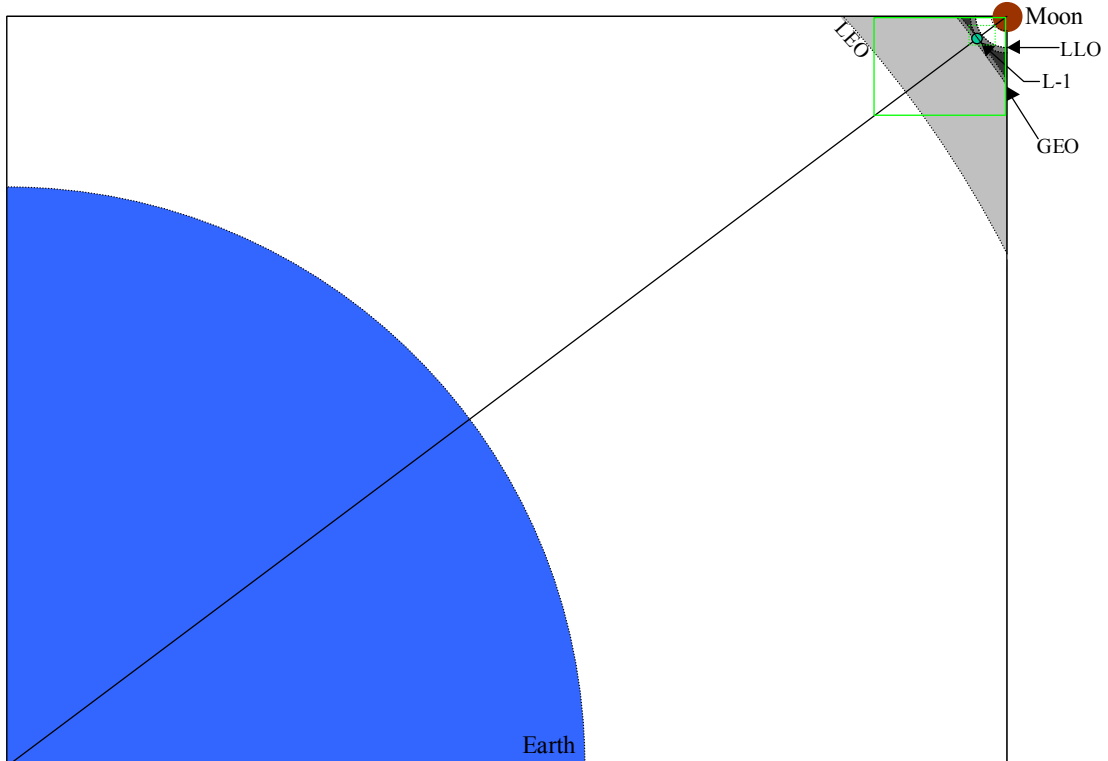
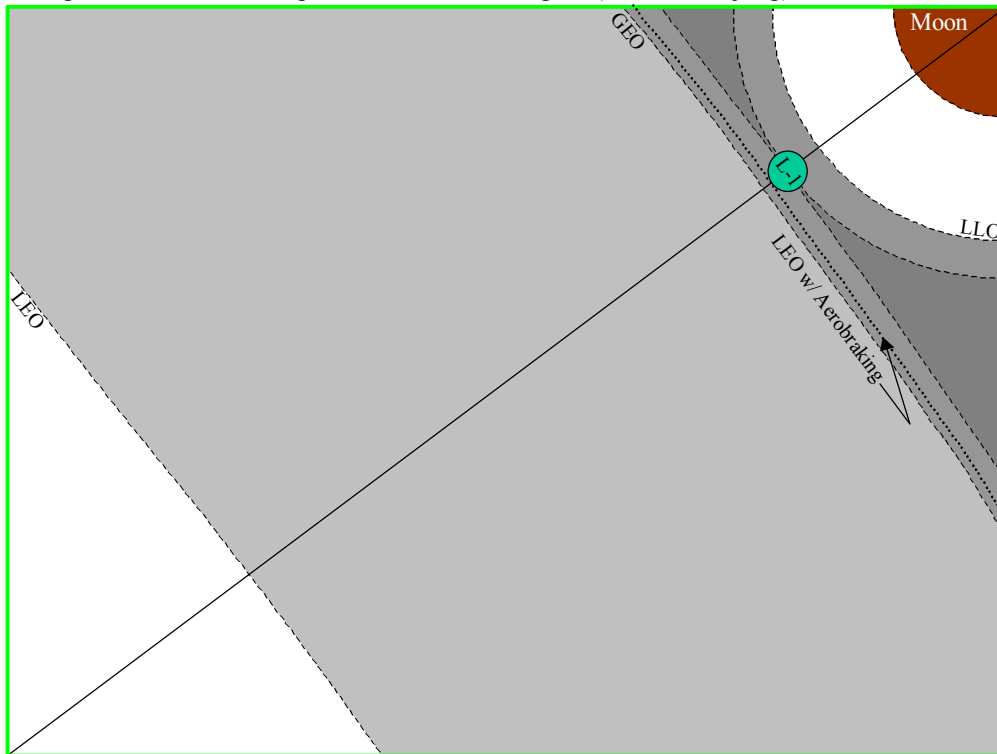


Figure 2.3. ΔV^2 close-up of the LEO-Moon region ($1'' = 4.25 \text{ MJ/Kg}$).



2.3 SPACE MINERAL RESOURCES

For many years, the possible presence of water ice in the lunar regolith has been one of the rationales used in the lunar science community to justify further lunar exploration, on the basis of its perceived value as a resource. The existence of permanently shadowed craters near both the lunar North and South poles was confirmed by the Clementine (see Nozette et al., 1995). Lunar Prospector data (see Feldman et al., 2001) demonstrates enrichment in the hydrogen concentrations in these polar regions, suggesting ice concentrations on the order of 1.5 weight percent of the regolith (i.e., one ton of lunar regolith may contain as much as 15 kilograms of water ice according to Neutron Spectrometer data). This value represents an average over a large area (the footprint of the Lunar Prospector Neutron Spectrometer instrument is a 60km arc – see Feldman, 2001), and the chance of higher ice concentrations is good. While the discovery of ice has increased the public perception that commercially significant resources may exist on the Moon, the demonstration of commercial feasibility is a more complex matter.

Other mineral-based resources also exist in space. Among those frequently cited are noble metals in stony iron and iron asteroids and lunar helium-3, both of which involve the extraction of trace constituents from regolith. The basis for considering these resources is that there is an identifiable demand on or around Earth. However, space resources will most likely be used in space. Therefore, it is likely that those that are most easily and reliably obtained will be used first. These could include water, wherever it is found, oxygen for propellant, metals and silicate minerals for construction or

manufacturing, silicon for solar cells, etc.

3.0 INTEGRATED MODELING APPROACH

The previous sections reviewed the rationale for considering a private venture producing lunar water-based propellant for use in Earth orbit. This section proposes a general integrated financial and engineering modeling approach to assess the financial viability of such a venture. This approach will be used in the following sections to conclude on the case study.

Multi-disciplinary science, engineering and financial inputs are required in order to model all relevant aspects of a private venture in space and bridge the gap between NASA and the private sector. An integrated financial and engineering model based on a private investor perspective is one way to bridge this gap, for three main reasons:

- First, an architecture optimized from an engineering point of view is not necessarily the most interesting for a private investor. For example, in the framework of a growing demand, economies of scale could lead the engineer to build up in the first year the capacity needed ten years down the line; while the private company might prefer investing in a scalable architecture, and build up capacity only as demand increases.
- Second, the metrics of interest to private sector investors differ from those that public sector engineers traditionally use for economic analyses. A ‘business case analysis’ is required to translate the engineering costs estimates into the metrics of interest to private sector investors.
- Third, an informed and effective public policy and strategy for space exploration demands that architecture trades, and initiatives regarding the private sector assess a wide range of scenarios. A single business case yields a specific outcome that is a function of its baseline assumptions. For NASA to effectively incorporate the private sector into its long-term plans, it should explore a wide range of potential space ventures, the conditions under which they would flourish, the steps that NASA can take to encourage them, and the public benefits/costs of those steps. To make these numerous case studies fast, accurate and comparable, a common analytic framework is required.

3.1 THE CASE FOR A PRIVATE INVESTMENT PERSPECTIVE

As a team of public-sector aerospace engineers designs a space architecture, the only economic information they typically compute are the architecture cost elements (development, production, launch, operations). Applying a government discount rate and adding up yearly costs yields the Net Present Value (NPV) metric they widely use to compare designs for commercially oriented missions. For example, to assess the potential of using Orbital Transfer Vehicles (OTVs) to transfer satellites from LEO to GEO, one would compare the lifecycle NPV of a GEO mission without OTV, to the lifecycle NPV of the same GEO mission with OTV. If the latter turns out to be more expensive, the venture is clearly not viable. If on the other hand the mission with OTV is cheaper, there might be a potential market for OTV transfer. Is that sufficient for private

companies to start investing in the venture? Unfortunately it is not, particularly in today's competitive capital markets.

Capital markets view commercial space as unpredictable, illiquid and high risk: high capital intensity, extensive R&D and regulatory costs translate into long and expensive product development cycles. Markets are often immature and unpredictable, or perceived to offer limited growth potential. Governments often subsidize competition, and market exit is difficult or 'sticky'. Shareholdings are illiquid and long term. Accordingly, any venture starts against significant financial impedance, and a simple NPV calculation does not give the information on which a private company would actually base its investment decision.

The first question asked by an investor is: "What are the discounted net present value and the effective rate of return on my equity investment?" Two common metrics used to answer this question are discounted Enterprise Value and discounted Price to Earnings multiple value in "Year X":

- Year X is defined in terms that an investor might be willing to endure – at most seven to ten years. If a venture cannot show interesting value in that timeframe, decision makers will turn to their other investment choices, especially in the framework of uncertain demand.
- The Enterprise Value (EV) is typically used when a company is privately held, and thus there is no public market valuation for the equity. EV in Year X is essentially the cumulative net value of the cash that the investors would achieve if they sold their stake in Year X.
- The discounted Price-Earnings (P:E) metric is used when the equity is publicly traded. P:E measures the value of the shares of stock as a multiple of the company's earnings per share. In essence, this valuation predicts what the shares will be worth in Year X, and thus provides a basis for calculating the real rate of return for the equity investor.

In both cases, the appropriate discount rate accounts not only for the effects of inflation, but also for the perceived risk of the venture: a dollar of return today is more predictable, and less risky than a dollar of return in the future. A decision to invest requires that the discounted future return on the investment not only be positive, but exceed an acceptable threshold, relative to the business' perceived level of risk and alternative uses of that capital.

If the rate of return for EV and/or P:E is sufficient, the private investor might then want to consider a "breakeven" analysis". Typically, this moves from top to bottom of an Income Statement: Gross margin breakeven (how soon can we make revenues greater than our direct costs of production?); EBITDA breakeven (how soon can we make revenues greater than our on-going cost of running the business?); EBIT breakeven (how soon can we make net revenues after accounting for the depreciation of our capital) and Net breakeven (how soon can we make money after paying the interest on our loans and taxes?). The financial attractiveness of a venture improves as these breakeven periods contract; conversely, as breakeven period lengthen, investors become less tolerant of risk and will impose a higher discount rate to account for uncertainties.

3.2 THE FINANCIAL MODEL

CSP Associates, Inc. developed a generic financial model to translate engineering cost numbers into the financial parameters just described. The tool models in a very generic way the three principal financial accounting documents that are used to calculate the performance of a private sector enterprise and yield the desired valuation metrics:

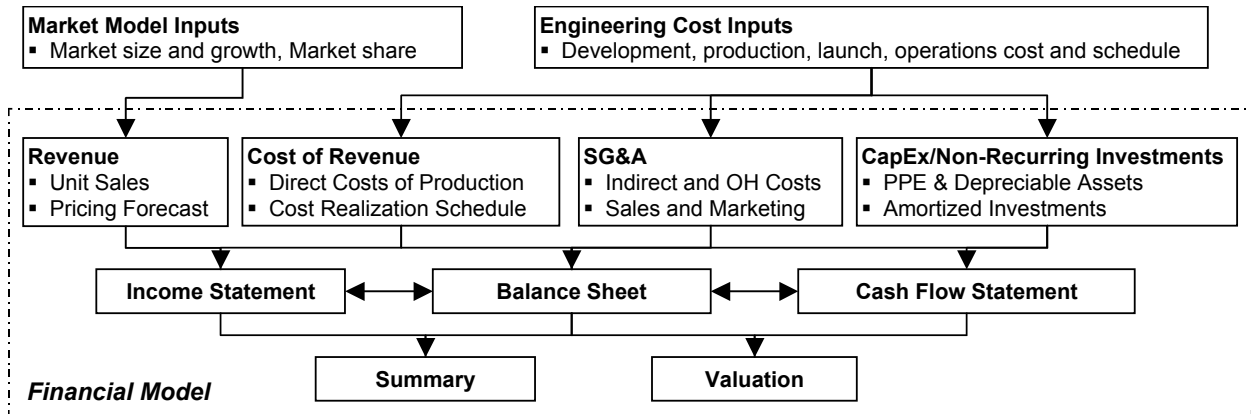
1. An Income Statement documents the profits and losses of the venture. Starting with the generated revenues, it subtracts first the cost of goods sold, then the sales, general and administrative costs (SG&A), the estimated depreciation and amortization, the debt interest payments, and calculates the taxes, to finally yield a net income.
2. A Balance Sheet provides an annual snapshot of the firm's year-end assets (sum of current assets such as cash and receivables, plus long-term assets such as the value of physical plant) versus its liabilities (sum of current payments owed by the company, long term debt, investor's equity and retained earning/losses).
3. A Cash Flow Statement characterizes the venture's cash flows, in other words, where the required funds come from (revenues and financing) and what they are used for (recurring and non-recurring expenses, financing costs). The statement incorporates assumptions on the firm's capital structure strategy, i.e. the proportion of debt and equity used for funding.

As illustrated in Fig. 3.1, these Pro-Forma statements require four types of financial inputs that in turn rely on outputs from the demand and engineering analyses:

1. The Revenue inputs require a quantitative estimate of demand as a function of time, in terms of quantity of demand (forecasted number of units of the product consumed each year), market share of the venture (percentage of this total product market captured by the venture each year), and unit price through time.
2. The Cost of Revenue inputs describe the direct marginal cost of producing each additional unit, each year. For a space venture, these typically include manufacturing, operations and delivery cost.
3. The SG&A (sales, general and administrative) inputs describe the indirect costs of business operations; this includes the costs associated with management, executive and marketing staff, staff training, overhead, rent, etc.
4. The CAPEX (capital expenditures) inputs require an estimate of all non-recurring investments and their amortization schedule; in the case of a space venture, this comprises all development costs as well as the cost of facilities and equipment, including all space elements.

These four types of required outputs lead the development of the integrated engineering and economic modeling approach.

Figure 3.1. Four primary input sheets drive the financial model.



3.3 INTEGRATING THE ENGINEERING AND ECONOMIC INPUT

This section describes the nine generalized analysis and modeling steps that can be applied to a candidate space resource case study to yield financial viability results. This modeling approach implies a constant interaction between the engineering and financial perspectives. At each point in the analysis, engineering factors (development and operations costs, schedule, performance, and risk assessments) have a direct impact on such issues as total investment requirements, the type and cost of financing likely to be used, the length of time to achieve positive cash flow, and venture operating margins and profitability. The nine steps are space resource definition, case study selection, demand modeling, engineering analysis, cost analysis, financial modeling, scenario optimization, sensitivity analysis, and conclusions.

3.3.1 SPACE RESOURCE DEFINITION

In the lunar propellant case study that will be studied, a raw resource from space (lunar water) is used by a private venture. However, the proposed modeling approach is not limited to space ventures that use material from space; serviced-based ventures such as on-orbit servicing or even remote sensing are very suited to the same approach. Even more that the availability of raw materials in space, what makes a space resource interesting from a financial viability standpoint is its potential for being of *direct* interest to customers. We will therefore use the following definition of a space resource:

A Space Resource:

- is a *Product* or *Service*
- has part of its supply chain and/or market in Space
- has a *direct* customer base on Earth or in Space
- is counted in units that reflect the *Pricing* structure that customers are willing to pay for the Resource.

For example in the case of lunar propellant, the space resource is defined as LEO-to-GEO transfer instead of water or propellant. It is counted in units of number of unit masses

transferred to reflect the pricing structure both of competition (launch from Earth directly into GEO) and supply (lunar propellant production).

3.3.2 CASE STUDY SELECTION

At the present time, few potentially viable private ventures for Space Resource development have been identified and most of them are associated with conjectural markets, such as those listed in the Commercial Space Transportation Study (CSTS). With improving technologies, the number of opportunities for private space ventures will increase as space activities expand and in time. The approach used here to model a space transportation business will be useable with other opportunities, such as recovering precious metals from asteroids for use on Earth, or transporting raw materials from the Moon to Earth orbit to construct solar power satellites. Which of these case studies has the most potential?

Even propellant based on lunar water, which requires relatively simple processing, still must be extracted, purified, and liquefied before the customer can be expected to buy it. A similar set of processes, in some cases including manufacturing to meet specific functional requirements, will be needed to bring any Space Resource to its economic use. All steps in the process that lead to the ability to sell the product must be included in the analysis, which must demonstrate sufficient effectiveness to meet the market price constraints. So in principle, case studies will be selected when there is some preliminary indication that the processes exist that can produce a Space Resource at less cost to the customer than competition. In most ventures, there will be at least one competitor: providing the Space Resource from Earth.

Early case study validation should be may be made at a high level, with back-of-the-envelope estimates of engineering and financial parameters. A number of case study ideas can be ruled out from the get-go by considering a series of necessary conditions for viability. These conditions start with the need for a market and for a clear advantage over competition, and go on with quick payback ratio analyses at various levels, such as:

- Is the venture likely to consume more of the Space Resource than it produces?
- Is the venture likely to require more mass to be launched to LEO than it will save in customer launch mass?
- Is the venture marginal cost of production likely to be smaller than the price customers are willing to pay?
- etc.

Ruling out bad ideas early can only help pinpoint the venture of most financial viability potential.

3.3.3 DEMAND MODELING

Once a case study has cleared a high-level technical and financial feasibility check, a more detailed business case can be developed. This starts with a market or demand model that yields three main outputs: total market demand and projected growth rates; the market share that the venture expects to capture, and the price at which the venture can

sell its product or service. Although the demand model will be specific to each case study, some general modeling rules apply to any commercial space market.

Annual market demand is the number of units of the product or service that are expected to be consumed each year. For example, several studies (CSTS, 1994; Smitherman, 2001) have forecasted the number of satellites to be launched as a function year, orbital regime, satellite type, and even satellite size. This type of analysis can be very useful starting points for any demand modeling. In addition, a thorough study should estimate the potential for new markets emerging from the availability of the space resource. For example, the availability of in-space refueling would see the emergence of new space missions such as maneuverable fleets of satellites.

Price forecast modeling involves an analysis of the maximum price each type of customer mission would be willing to pay for the space resource. For existing markets, the product or service must provide an advantage over the current way of doing business; quantification of this benefit readily provides an upper bound on the price that can be charged. For example, the price for “LEO-to-GEO transfer using lunar propellant” must first cost less than a traditional ELV or Shuttle staged launch to GEO, and second cost less than an OTV using Earth-based propellant. Similarly, the price for on-orbit servicing must be cheaper than satellite replacement, but also than designing a spacecraft with a longer mean mission duration. For potential new markets, a more involved analysis is required to estimate the maximum price that will allow the market to emerge; nested “private ventures in space” analyses might be required if the new market is itself a space venture (e.g. at what price of ‘Commercial Service X’ does ‘Commercial Venture Y’ become feasible?)

Finally, **market share growth** accounts for the rate at which the potential customers actually turn to the venture. This depends on several factors, such as the number of competitors, market differentiation, and customer perceptions of risk/confidence. As a necessarily highly uncertain parameter, market share growth is an important candidate for sensitivity analysis.

3.3.4 ENGINEERING ANALYSIS

The engineering model, or architecture design, combines the minimum set of system elements required to effectively deliver the Space Resource to its market. Although this design will be case-specific, a few simple rules of thumb apply:

1. **Focus on timelines and cost.** The venture expenses and its times to break-even are key to the financial viability of the venture. The basic model must capture technology, deployment, production, launch, and operational considerations with just enough definition to estimate timelines and costs.
2. **Favor scaling laws over point designs.** Rather than a static point design, what is helpful is a more general engineering model or tool that can accommodate a range of starting assumptions and their associated cost factors. Database-linked or analytical engineering scaling laws for example provide flexibility to meet the modeled demand. For example, the engineering model developed for the lunar propellant case study defined a unit-size architecture designed to meet a small amount of demand, and launches incremental units as demand increases. Beyond engineering scalability, this

approach has the advantage of decreasing the risk associated with uncertain demand growth.

3. **Start with a simple model** The same modeling approach applies to any level of detail, with the quality of the financial viability results depending only on the quality of the inputs. Starting with a very high level model can help carry out simple trade studies and identify scenarios that are worthwhile taking to the next level of detail. Preliminary modeling can begin with a technology list and mass breakdown for each primary system, while successive iterations will evolve more advanced technical descriptions for nested subsystem elements. The initial set of inputs defines the 'baseline scenario.'

3.3.5 COST ANALYSIS

The cost model must correctly anticipate technology level, design, development, production, launch, operations and maintenance costs. In addition, it must be scalable to adapt to scaling designs. Although not as accurate as grass roots or analogy-based estimates, cost models based on analytical Cost Estimating Relationships (CERs) are ideal for this application. CERs provide an estimate of cost and cost uncertainty based on a number of high-level engineering parameters that are readily available from the engineering model (such as type, mass and technology readiness level of each subsystem). This provides not only the required flexibility to quickly adapt to changing designs, but also the required inputs for cost risk analysis.

Similarly to engineering model, cost models can be developed at various levels of detail. For a first round of analysis, the cost model could be as simple as CERs based on total dry mass for development and production cost, wet mass for launch cost, and number of elements for operations cost.

It should be noted that the CERs typically available are derived primarily from government programs. It is conjectured, as space industrialization grows and technology becomes better understood, more reliable and more widely used, and particularly as commercial incentive structures replace government contracts, that costs could drop well below than those shown in current cost models. This is particularly true for mass production, and any private venture cost model should include learning curve effects.

3.3.6 FINANCIAL FEASIBILITY

The cost, performance and schedule outputs become inputs to the financial model, creating the initial assessment of financial viability. If the venture is not viable the financial model shows the main cost drivers, which in turn can be used to explore either alternative technologies or architectures, or to explore different versions of the baseline scenario by changing the primary assumptions or technologies. In addition, as all production processes used in space will have a startup cost and operational overhead, it will be desirable to know at what scale of production (and demand) the case study can be profitable.

3.3.7 FEEDBACK AND SCENARIO OPTIMIZATION

Preliminary investigation of the integrated model may identify areas in which the scenario may be improved. This can be done at the level of the cost model for the engineering system, in which the high-cost elements of the architecture can be analyzed and solutions found to reduce the scale or even eliminate an element of the architecture. For example, initial examination of the lunar ice architecture incorporated an all-propulsive approach to the transportation system, in which lunar propellants were utilized throughout. It was quickly determined that these architectures were economically infeasible, and architectures that involved aerobraking to low Earth orbit were introduced.

Analysis of the financial viability results from the first round of modeling can help guide refinements in engineering and financial assumptions. The use of new technologies, which have the potential to reduce key mass and cost drivers, can be traded against the additional cost and time associated with their development and validation. The impact of pricing strategy can be tested. The possible government incentives to release key hurdles can be identified.

The goal of this analysis step is to identify a scenario that combines realistic market assumptions, an efficient and feasible architecture design to meet this market, realistic cost estimates, and reasonable assumptions on government participation, into a close-to-financially viable private space venture.

3.3.8 SENSITIVITY ANALYSIS

Another tool that can be utilized is sensitivity analysis. This can be applied at the component level, such as comparing alternative ways of providing power for lunar surface systems. The sensitivity of the economic results to the grade of the resource will be important, as was shown in the lunar ice case. The model's economic assumptions can also be studied. For example, discount rates or the degree of sharing of public/private investment can be modeled. Finally, market assumptions can be tested, such as the size of the market or acceptable costs for the resource.

Once a good scenario has been identified, sensitivity analysis is key to test the impact of uncertain parameters and analyze the conditions for financial viability. A key metric to plot is the rate of return on investment in Year 10 for private investors in Year 1: this rate must exceed a given threshold to private investors to be interested (typically above 20% for risky ventures). Key parameters to test include (but are not restricted to):

- Market demand and market share growth. These parameters are typically very uncertain in any space venture, especially if the venture is launching a new product or service. The minimum demand and demand growth required for the venture to be viable can be compared with the expectations and their uncertainty.
- Discount factors. Private investors use discount rates to account for the perceived risk of the venture. The higher the perceived uncertainty, the higher the discount rate and the required return on investment. Since risk is always hard to quantify, it is important that the venture be viable for a range of discount rates.

- Launch cost from Earth. Whether a provider of service to the venture or a competitor, launch from Earth is bound to be a key player in any space venture's financial viability. The sensitivity of the venture to launch costs is particularly interesting as these costs are expected to drop in the coming decades.
- Key technological parameters. Testing the sensitivity of the venture to parameters such as propulsion system performance, specific masses of various components, or specific power of power sources, can help identify the key technical drivers and the areas of most interesting potential for technology development.
- Alternate government incentives, such as participation in development costs, tax rate, or guaranteed price and/or customer base. Another way to assess government incentives impacts is to assume the availability of technologies and/or assets in space at the start of the venture, which might reduce timelines and cost. This analysis can help identify the most efficient government incentives to foster private sector involvement.

3.3.9 CONCLUSIONS

After analysis using any or all of the above tools, a case can be made (or refuted) that a particular resource is economically viable for a particular market. It will be important at this stage to fully document all assumptions so that the reviewer can gauge the completeness and quality of the analysis.

The results of the sensitivity analyses described above can help draw a map of the conditions for financial viability of the space venture. The capabilities offered by such a modeling approach and the type of conclusions that can be drawn will be illustrated in the following sections on the lunar propellant case study.

Applying the same approach to a number of private venture cases studies can help draw a general map of the respective roles for the private sector and for the government in future solar system exploration.

4.0 CASE 1: LUNAR PROPELLANT FOR LEO-GEO TRANSFER

At the present time, few commercial activities that utilize known space resources have been identified, and none have been shown to be commercially feasible. The CSTS (1996) report and others suggest that water extracted from space resources may be the best candidate for commercial activity because of an existing market. The particular market or need that is identified for Case 1 is in-space transportation, specifically LEO-GEO transfer. The possibility that the need might be met using lunar resources, rather than bringing the required materials from Earth, has been extensively studied. Cryogenic propellant, as studied in the current model, requires relatively simple processing and water ice has been demonstrated to exist at both lunar poles. Steps in the process from resource extraction to utilization are described below.

The approach used here to model a space transportation business, specifically selling transportation services provided by an Orbital Transfer Vehicle (OTV). While others have demonstrated technical feasibility using the “mass payback” criteria (which compares the mass of equipment and propellant for production and delivery of the product to its place of use to the mass of the product that would have to be delivered from Earth - see Stancati, 1999; Rice, 2000), a positive mass-payback relationship does not guarantee an economic benefit. The approach used here is to step beyond the engineering modeling and create a foundation to show commercial benefits.

4.1 THE CASE 1 ENGINEERING MODEL

In the Case 1 model, lunar regolith is mined, the water removed by raising its temperature and condensing the water that is evolved, then the water electrolyzed and the product liquefied to produce liquid hydrogen (LH₂) and liquid oxygen (LOX) for propellant. A reusable space tanker that uses lunar LH₂/LOX transfers water to a space propellant depot at the Earth-Moon L1 point. This location has the advantage that it is always in the same position with respect to the Moon, providing anytime access back and forth, and is similarly placed with respect to Earth.

To correctly anticipate technology level, design and development factors, the Case 1 model limited itself to systems that have heritage (i.e., proven technologies). A spacecraft launched from L1 can enter any Earth orbit using about the same energy, although the trip must be timed properly to rendezvous with an object already in Earth orbit. Once the water has been delivered to L1 there are several options. In this study, two different options were considered. In the first option (Architecture 1, figure 4.1), a second propellant depot is established in LEO, presumably in equatorial orbit. Water is transferred from L1 to LEO, electrolyzed and liquefied in LEO, and used to fuel a reusable orbital transfer vehicle that transports payloads from LEO to GEO. The OTV then returns to LEO for refueling. In the second option (Architecture 2, figure 4.2), the reusable orbital transfer vehicle operates from L1, flies to LEO to rendezvous with a payload, takes the payload to GEO, then returns to L1 for fueling and another trip. This eliminates the need for a second refueling depot. Note that this section of the report is intended to present an overview of the development of the models for Case 1 Architectures 1 and 2. Details regarding specific technical assumptions can be found in Appendices 1-3.

Figure 4.1. Architecture 1 for Transporting Payloads from LEO-GEO Based on Lunar Propellants
(Note: ΔV^2 close-up, scale: 1" = 4.25 MJ/Kg).

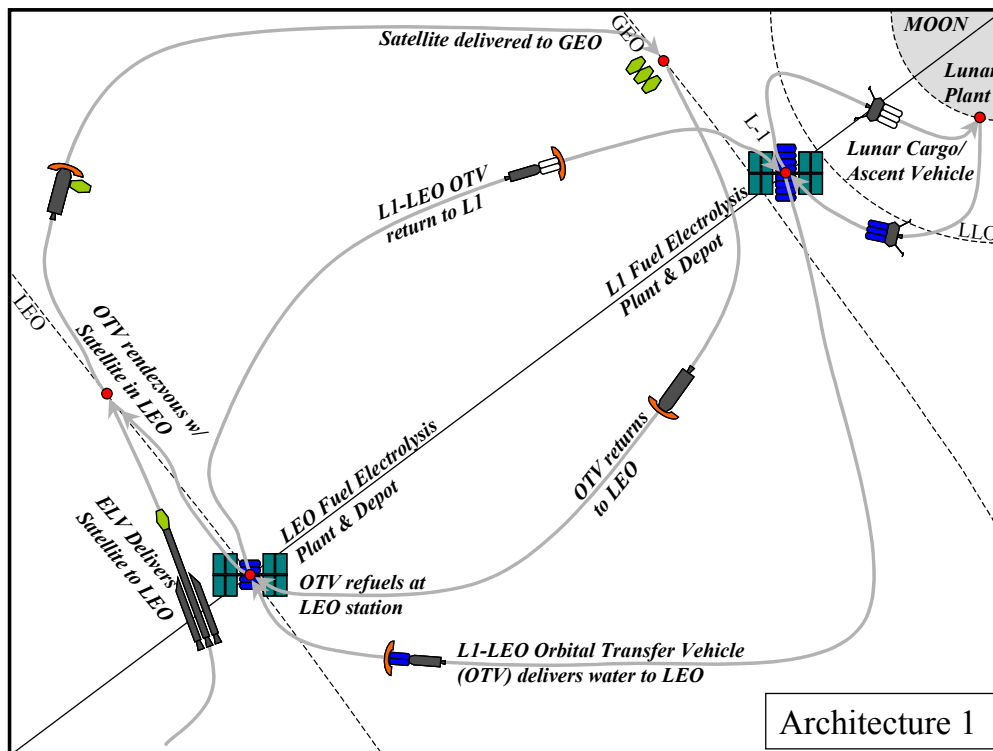
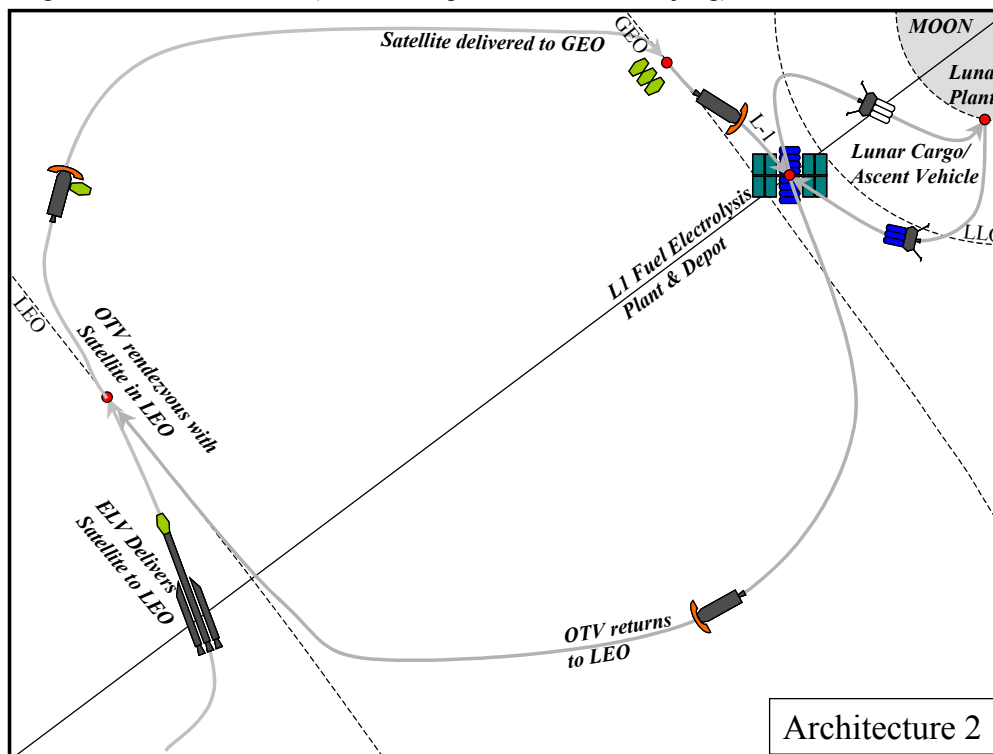


Figure 4.2. Architecture 2 (ΔV^2 close-up, scale: 1" = 4.25 MJ/Kg).



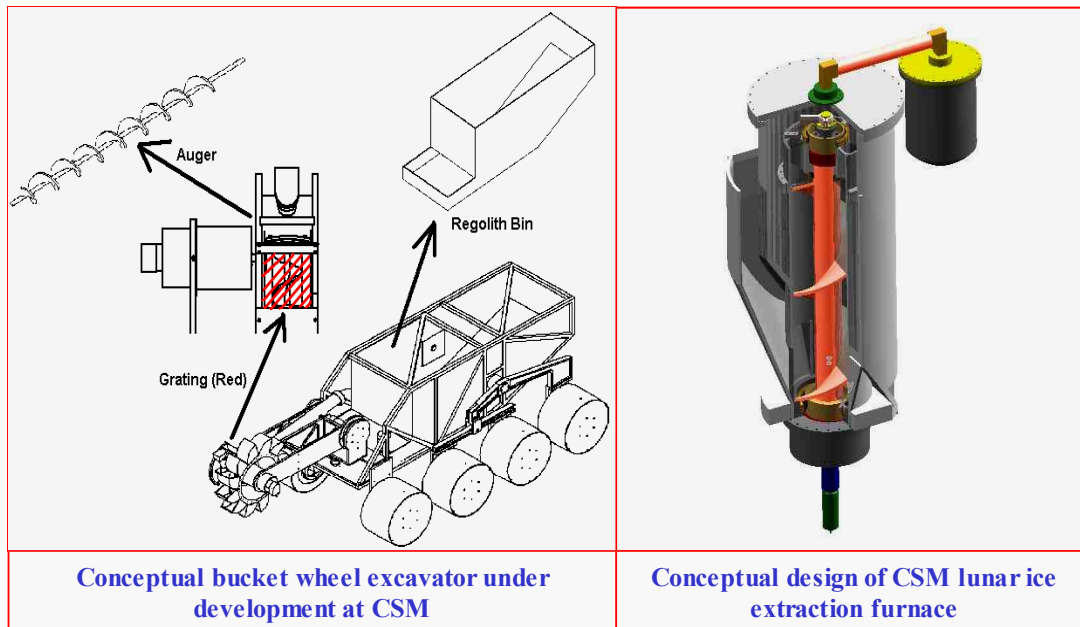
4.1.1 MINING AND PROCESSING SYSTEMS

The lunar surface mining and processing system consists of equipment to mine regolith, extract its water, electrolyze the water to produce gaseous hydrogen and oxygen, liquefaction equipment to liquefy the gases, and a storage capacity. Power must be provided for the facility. The surface system also must include a launch/landing facility with the capability of transferring the payload (water) and propellants (LOX, LH₂) to a tanker that will transport water to L1.

A baseline conservative assumption in the model is that the regolith contains 1% water by weight (note that the estimated value by Lunar Prospector is ~1.6%). It is assumed that all water and propellant production is carried out within the permanent shadow, although other options for the lunar system exist (see Duke et al, 1998) and should be investigated in further studies. A nuclear reactor is assumed to provide thermal and electrical energy for water extraction. The system extracts water by heating regolith from its ambient temperature (80K) to 200K under vacuum. Water is electrolyzed and the hydrogen and oxygen liquefied and stored for propellant. Liquid oxygen can be stored using passive thermal control techniques in the permanent shadow and the energy cost of storing liquid hydrogen is minimal. Water tanks must be insulated and heated to retain water in liquid form. The “specific mass” or “specific energy,” which are defined as the mass or energy required to produce a given amount of product in a given amount of time, are provided for the major elements of the surface architecture in Table A1-4, along with other general assumptions utilized in the model. For costing purposes, it is assumed that 10 % of the system must be replaced each year of operations. The current architecture assumes that excess oxygen, which appears because the oxygen content of water is higher than that of the fuel mixture used in LH₂/LOX rockets, is lost to the system. Enough hydrogen and oxygen are stored on site to allow for continued operation of the system when a lunar water tanker is not present at the production facility. Otherwise, the product is stored in the tanker itself.

Data for the various elements of the lunar surface system have been extracted from Eagle Engineering (1988). Some of these data, particularly for excavation and extraction systems, have been under study at the Colorado School of Mines. A bucket wheel excavator modeling (Figure 4.3) has demonstrated the potential to excavate as much as 10 times the system’s mass per hour. Therefore, excavator mass is not considered a major driver for the total plant mass on the lunar surface. Extraction systems tend to be more massive, with calculations suggesting that a system can process its own mass of regolith in one hour (the energy required to heat the regolith is modest, due to the low vapor pressure of water in vacuum). Indeed, the majority of electrical energy consumption is used to electrolyze the water and liquefy the propellants. Nuclear systems have been assumed in the current model; however, options exist for the use of solar energy, which can be collected in areas adjacent to the shadowed craters (where sunlit areas can be found more than 80% of the month). Solar systems will be less massive and less costly than nuclear systems, although the issue of intermittent power availability and channeling the energy from near-permanent sunlight into a permanently shadowed crater can create certain design complexities. Details regarding the complete list of assumptions for the lunar plant can be found in Appendix 1 and 2.

Figure 4.3. Mining and extraction systems under development at CSM.



4.1.2 TRANSPORTATION ARCHITECTURES

The two architectures as depicted in Figures 4.2.1 and 4.2.2 have similar space transportation systems. In option 1, a reusable lunar tanker, which can land repeatedly at the production site on the Moon, is fueled with LH_2 and LOX , and carries a payload of water to L1. At the L1 depot, water is converted to propellant needed to return the lunar water tanker to the Moon and to send a separate water tanker spacecraft to a depot in LEO. This vehicle is reusable and flies to LEO using an aerobrake. At the LEO depot, the remaining water is converted to propellant and stored for delivery to reusable orbital transfer vehicles that deliver satellites from LEO to GEO, coming back empty to LEO using an aerobrake. A portion of the propellant also is utilized to return the water tanker to L1. Values for ΔV that are used in both architectures are reported in Table 4.1.

Table 4.1. ΔV assumptions used in transportation system modeling.

LEO-GEO	3800m/sec
GEO-LEO with aerobraking	500m/sec
GEO-L1 (assumption only)	800m/sec
L1-LEO with aerobraking	500m/sec
LEO-L1	3150m/sec
L1-Moon's surface	2390m/sec

An assumption is made that all vehicles use liquid oxygen and hydrogen fuel, with an Isp of 460 and a mixture ratio of 6.5:1. This mixture ratio is a matter for propellant system design, but perhaps represents a reasonable mixture for a highly reusable propulsion

system, although it is a little higher than currently utilized in the Space Shuttle's main engines. Because the ratio is not stoichiometric for water, anywhere in the system that propellant is produced from water, excess oxygen is created. This excess oxygen is given no commercial value in our current models, although it is produced on the lunar surface, at L1 and, in the scenario for Architecture1, in LEO.

In addition to the production plant, both architectures share common elements as described below (although sizes differ slightly - detailed design parameters and vehicle sizes can be found in Appendix 1 and 2):

L1 Propellant Depot: At this depot, water is received and propellant is produced. The electrolysis and liquefaction systems are similar to those on the Moon. Excess oxygen produced at L1 is assumed lost to the system. Power for the propellant depot is provided by solar arrays.

Lunar Water Tanker. This vehicle is capable of landing near the propellant production plant (probably not in the permanent shadow), taking on a payload of water and cryogenic propellants and traveling from the Moon to the L1 propellant depot. The tanker is assumed to be highly reusable, with 10% per year hardware refurbishment.

In addition, each architecture has distinct elements. For Architecture 1, these include (detailed design parameters can be found in Appendix 1):

L1 to LEO Tanker. This system has an aerobrake for entry to LEO, assuming a mass fraction of 15% of the mass entering LEO (spacecraft and water payload). As Architecture 1 includes a depot in LEO, the tanker is sized for refueling in LEO. The LEO propellant depot is sized using the same assumptions as those for the L1 depot.

LEO Propellant Depot. At this depot, water is received from the L1 depot and propellant is produced for the LEO-GEO-LEO or LEO- L1 transfer system. The electrolysis and liquefaction systems are similar to those on the Moon. Excess oxygen produced at LEO is assumed lost to the system. Power for the propellant depot is provided by solar arrays.

LEO-GEO-LEO Orbital Transfer Vehicle: OTV for carrying payloads from LEO to GEO and returning to LEO with an aerobrake. This vehicle's mass is estimated using the same performance parameters ascribed to other tanker vehicles.

For Architecture 2, this includes (see Appendix 2 for detailed design parameters):

L1-LEO-GEO-L1 Orbital Transfer Vehicle. The transfer vehicle and tanker functions are combined and a single vehicle is fueled at L1, flies with a propellant load that is aerobraked into LEO where it performs a rendezvous maneuver with a satellite, then propels the satellite to GEO. Following the insertion, the vehicle flies back to L1 for refueling. This vehicle must carry to LEO the propellant needed for LEO-GEO-L1 as well as the aerobrake for entering LEO.

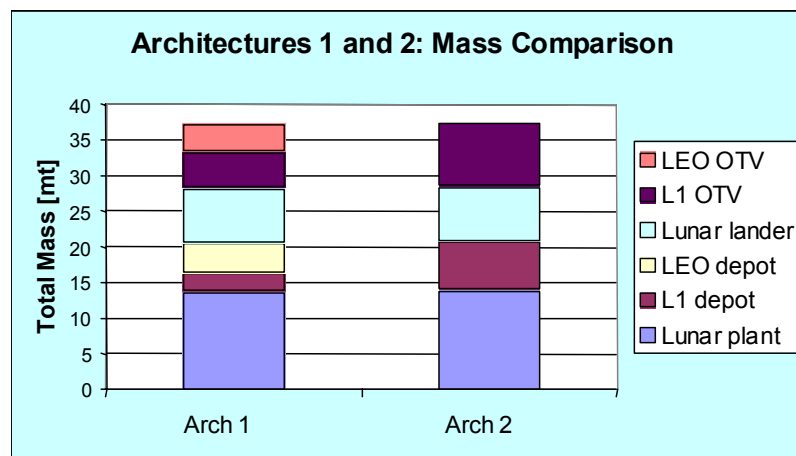
Architecture 1 has the advantage that the delivery of propellant from the LEO depot can be metered to the user in proportion to the needs of the LEO-GEO mission. Additional customers could be served by increasing the rate of water delivery to LEO with the water tankers and by increasing the propellant production capacity at the depot. The propellant

depot would have to be in a fixed orbit and might not be suitable for the fueling of certain satellites; for example, a depot in equatorial orbit may not be amenable to fueling a mission requiring polar orbit. This architecture would allow the entity that delivers satellites to GEO to be a separate business from the one that produces propellant in LEO and from the entities that produce water on the Moon and transport it to LEO.

In Architecture 2, it is assumed that the only propellant depot is located at L1. For delivery of a satellite from LEO to GEO, an OTV is fueled at L1, aerobrakes to LEO, docks with the satellite that is to be delivered to GEO, flies to GEO and then returns to L1 for its next mission. This is similar to an architecture studied by Sercel et al (1999). The LEO propellant depot and the separate water tanker from L1 to LEO are not required. Because the OTV cannot be refueled in LEO, it must carry with it from L1 the propellant needed to get from LEO-GEO-L1. This architecture, as well as being somewhat simpler (it eliminates a LEO propellant depot and one of the types of OTV), has the advantage of allowing access to a variety of different Earth orbit inclinations with similar propellant requirements from L1. It is also amenable to an integrated business plan, in which the lunar mining and space transportation functions are provided by a single entity.

Sizing for each of the architectures starts with the total consumption of propellant per period. This is calculated by combining the transportation system assumptions (each vehicle uses propellant during its operation cycle) with annual market demand. Working back through the system, the amount of propellant required at each location in the system and the mass of lunar surface systems is calculated. Then, the specific mass and power data of each of the elements is used to determine the mass of hardware required at each location. For each additional increment of capability, a similar increment of hardware is added. The amount of propellant used at each node is shown in Appendices 1 and 2 for the unit plant size (note: ten production units are deployed the final year in each model for optimal phasing with market capture).

Figure 4.4. Mass comparison of architectures 1 and 2.



4.2 THE CASE 1 ECONOMIC MODEL

The economic model is affected by many assumptions outside the architectural assumptions. These include assumptions such as: (1) the technology level assumed at the start of development – The NASA-Air Force Cost Model (NAFCOM) assumes totally new development of all elements; (2) whether the commercial investor must pay for the development costs - as many of the systems are common to other space activities, the financial model assumes that DDT&E costs are absorbed by other government programs in Versions 2-4; (3) the size of the market - efficiencies are gained as the number of units that must be produced increases; And (4) primary business assumptions - such as the rate of market capture, expected rates of return to investors, discount rates and taxation.

While the launch energy from the Moon to LEO can be as low as 4% of that required from Earth (see Section 2.2), the question remains as to whether such a system can save the final customer money and still produce enough profit to reward investors.

4.2.1 CASE 1 COST MODELING

The NASA and Air Force Cost Model (NAFCOM99) cost estimation tool was utilized to estimate the costs of development and production at the systems level for elements of each of the architectures. The masses derived from the architecture analysis were input into NAFCOM along with analogies appropriate for the current level of analysis. Cost summaries for the two architectures are shown in Table 4. NAFCOM generates the cost estimates for design and development (D&D), system test hardware (STH), flight unit (FU) and production units (Prod). The cost model is responsive to the unit mass of the elements utilized. In general, smaller units cost more per kilogram than a single large unit. However, producing a series of small units generates up front savings due to a much lower design and development cost (which can be as high as four times the production cost) that is incurred only once. This is a significant consideration for the case where a number of identical systems are installed over time (to incrementally expand capacity) because it decreases up front capital expenditures. For the current studies, no attempt was made to optimize the size of individual elements, although a 'learning curve' was applied to the costing of multiple units. Note that the NAFCOM99 modeling approach, which was used for the current study, requires that separate estimates be built for each of the modeled scenarios, limiting the flexibility of sensitivity analysis.

Added to the hardware cost are launch costs, as each of the hardware components of the system must be delivered to its place of use. The model assumes that the cost of transportation from Earth to LEO is \$10,000/kg (approximating the current launch costs of the Space Shuttle or Ariane 5). Transportation costs for initial delivery of payloads to L1 and the Moon are estimated as \$35,000/kg and \$90,000/kg respectively. Note that much of this cost is associated with transportation of propellant from the Earth into space.

Once the first propellant production unit is emplaced on the Moon, utilizing the lunar propellant and transportation vehicles reduces the cost of transportation of subsequent units of production. In our preliminary model, we assume that the first production unit, equivalent to that required to transport 3-5000 kg payloads a year from LEO to GEO, is installed and thereafter is utilized to provide propellant and transportation for new hardware for system expansion. After its installation, the cost of transportation of new hardware to L1 and the Moon is assumed to be equal to the cost of transportation to LEO,

as system elements are ‘handed to’ OTVs operated by the enterprise. The model currently makes the following assumptions regarding the delivery of additional units to destinations beyond LEO:

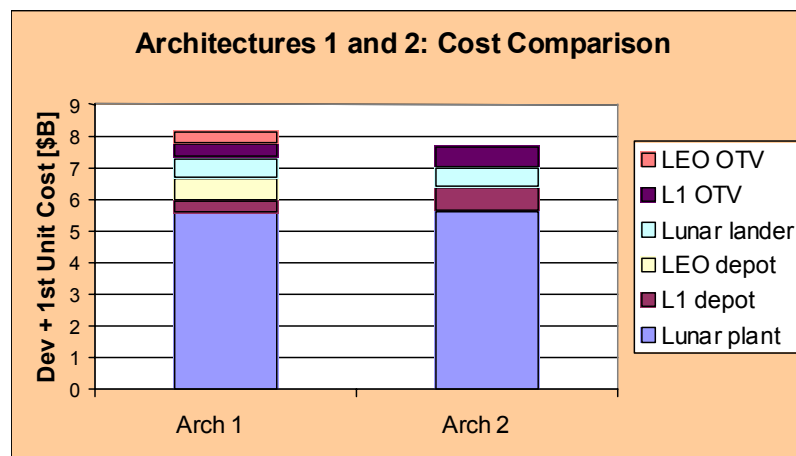
- Each year, the units required in the following year are launched
- For architecture 1, one unit is able to build up to 240 mt/yr (12 trips of the L1-LEO-L1 OTV and of the lunar lander, at 20 mt a trip)
- For architecture 2, one unit is able to build up to 117 mt/yr (12 trips of the OTV and at 9.7 mt a trip)
- Each vehicle can make up to 12 trips a year (for example in architecture 1, if demand increases from 15 mt/yr to 30 mt/yr, an additional lunar plant is required, but no additional LEO-L1-LEO OTV)

This assumption could be fine tuned by calculating the amount of propellant and vehicles in excess of the new hardware transportation requirement and incorporating the revenue from its sale into the economic analysis.

Operations costs are an important factor in commercial viability, and are modeled at the systems level using the Space Operations Cost Model (SOCM). It is assumed that all maintenance and repair is carried out by robotic systems and that on-site humans are not required for successful operation of the system. In fact, a one-ton maintenance facility is built into the architecture and cost estimates as part of each unit production plant deployed on the lunar surface. At a given level of activity, on-site humans for maintenance and repair may become economically desirable. However, circumstances under which that might become an effective approach have not been analyzed.

Finally, the economic model assumes that 10% of subsystems (and 1% of tanks) must be replaced each year. This provides a measure of replacement costs that is directly related to the mass of hardware being utilized in the system. The cost of producing the replacement hardware is scaled from the original NAFCOM estimates, and the transportation cost for the replacement hardware is included. Note that this places an increasing burden on later production years, comprising almost 1/3 of the capital cost in year 7 of system operation.

Figure 4.5. Cost comparison of architectures 1 and 2.



4.2.2 CASE 1 MARKET MODELING

Annual market demand is the number of units of the product or service consumed each year. Several studies have forecasted the number of satellites to be launched as a function year, orbital regime, satellite type, and even satellite size (CSTS, 1994; Smitherman, 2001). The launch of satellites from Earth to geosynchronous orbits (GEO) is an established and growing business. Each year, between 25 and 30 satellites are launched at a typical cost of \$35,000 per kilogram of satellite. This comprises the primary candidate market for the case under examination. For purpose of early analysis, both architectures assumed that a constant number of satellites must be delivered from LEO to GEO annually. The 'unit system' is sized to provide the capacity to deliver 3 x 5000 kg satellites from LEO to GEO (for a total unit capacity of 15,000 kg). One unit is deployed in the first year, building to 10 in six years for a total capacity of 30 satellites per year (150 tons of total satellite mass). This simple satellite demand model is derived by combining the 2002 GSO launch forecast (AST, 2002) and a satellite mass growth model (AST, 1999) into market projections for the period of 2010-2016.

The price model is assumed to have an upper bound - taken as the minimum of a traditional ELV launch to GEO vs. an OTV using Earth-based propellant (both systems are considered to compete with lunar-based fuel supply). Demand is priced as a function of satellite mass (dollars per kilogram transported). Note that the demand model based on Smitherman (2001, developed for in-space water-based propellants provided from Earth) lacks the cost estimates or economic data required to derive a competing price. Because the main advantage for customers is savings in Earth launch cost, the maximum price that can be charged is the difference between the cost to launch to GEO (\$35,000/kg) and the cost to launch to LEO (\$10,000/kg), netting to \$25,000 per kilogram of delivered satellite. A 20% 'discount' was assumed to be attractive to current customers, forming a simplified price function at a constant \$20,000/kg. A market capture function was added to the model, starting with 10% market share in the first operational year, and ramping up to 100% after 7 years of successful operations. Market share growth accounts for the rate at which the potential customers actually turn to the venture. This can depend on several factors, such as the number of competitors, market differentiation, and customer perceptions of risk/confidence.

Consider a recent example of a 4,460 kg payload launched to GEO by an Ariane 44L. As it passed through LEO, the cryogenic third stage required more than twice as much propellant (11,900kg) as the final payload (see Figure 4.6). Besides highlighting the leverage that lunar resources hold over their terrestrial counterpart, the potential exists to refuel and reuse the upper stage for additional satellite delivery (or other uses such as satellite servicing). The existence of a fuel depot within reach of the empty vehicle (at L1) creates a commercial incentive for the owner of the vehicle to find additional customers, stimulating space commerce.

Figure 4.6. Performance specifications for Flight 113 of the Ariane 44L launch vehicle.

ELEMENT	MASS (kg)	
	DETAIL	RUNNING TOTAL
AfriStar	2 739	
GE-5	1 720	
• Adapter + SPELDA + Vehicle Equipment Bay (VEB) + residual fluids & performance reserve	1 137	
• 3rd stage dry mass	1 241	
<i>Mass after end of 3rd stage propulsion</i>		6 836
• 3rd stage propellant	11 720	
<i>Mass after 2nd stage separation</i>		18 556
• 2nd stage dry mass + 2/3 inter-stage	3 496	
• Fairing mass	745	
• 2nd stage propellants	35 453	
<i>Mass after 1st stage separation</i>		58 250
• 1st stage dry mass + 1/2 inter-stage	17 995	
• 1st stage propellants	231 173	
• Liquid strap-on boosters (dry mass + propellants)	174 655	
Total mass at liftoff		482 073

4.3 CASE 1 MODEL RESULTS

This section presents results obtained from the integrated modeling of Case 1, Architectures 1 and 2. Consistent assumptions are used in the baseline model for both architectures (the baseline has been labeled ‘Version 0’). These assumptions are the most conservative, and are considered to be the most realistic – e.g., ‘current’ technology, ‘standard’ procurement and management, ‘normal’ investor behavior, etc. Iteration of the model to a financially feasible solution involved a process of changing progressive relaxation of assumptions (see Versions 1-5 in Table 4.2, below). The resulting feasible model (Version 5) indicates one set of possible conditions under which a commercial venture ‘could’ be profitable to private investors. The ‘realism’ of these feasibility conditions has been a subject of debate among team members. Note that while certain assumptions might be considered simplistic, and certain factors omitted (such as risk), the results are a good illustration of the analytic capabilities offered by the integrated modeling tool developed for this study.

4.3.1 BASELINE MODEL RESULTS

The baseline numerical assumptions for the case study included conservative demand, mass and cost estimates, and no government incentives apart from generic technology development. With these assumptions, the project Net Present Value (NPV) was quite negative (minus \$5 Billion), as shown in Table 4.3.

4.3.2 MODEL VERSIONS: FINDING A FEASIBLE SOLUTION

Use of the integrated modeling tool makes it possible to explore in real time the conditions for financial viability. As an example, Table 4.2 identifies the manner in which the model was adjusted in each of the versions. Table 4.3 summarizes the results

of several versions with incrementally ‘less conservative’ assumptions (for reference, the table also cites the traditional metrics: NPV and NPV-based rate of return). Table 4.4 shows the financial statements for the most feasible version for each of the architectures (Note: all model version calculations incorporate a discount rate of 10%, a cost of debt of 12% and an income tax rate of 40%). These results show how the model allows quick “what if” studies:

- What if the government pays for the upfront development and first unit costs? (Version 1)
- What if, in addition, the efficiency of commercial production reduces costs by 30% compared to the traditional NASA development and procurement approach? (Version 2)
- What if, in addition, the concentration of H₂O in lunar regolith is twice our baseline (2% instead of 1%)? (Version 3)
- What if, in addition, the demand for LEO-to-GEO transport is twice as high as our conservative forecast? (Version 4)
- What if, in addition, the price charged for orbital transfer (\$20,000/kg) is raised by 25% (to \$25,000/kg) (Version 5)

Version 4 yields a venture with positive NPV, but the investor’s return on equity (15.2%) is probably still insufficient to trigger investment (i.e. investors could probably achieve a similar rate of return in a more traditional investment). Therefore, Version 5 is considered to be the only version that achieves financial feasibility.

Table 4.2. Model versions relative to baseline.

Version	Description	Summary
0	Architecture 1&2 Baseline. All assumptions set to most conservative level.	Baseline
1	Baseline w/ No Non-Recurring Investments. (assumes that the public sector pays for design, development and first unit cost)	Remove DDT&E from Baseline
2	No Non-Rec. Investments + Reduce the production cost of all elements by 30%.	Add 30% Production Cost Reduction
3	No Non-Rec. Investments + Reduced production cost + Increase concentration of Water in Lunar Regolith from 1% to 2%.	Add 2x Lunar Water Concentration
4	No Non-Rec. Investments + Reduced production cost + Increase concentration of Water in Lunar Regolith + Double demand.	Add 2x Demand
5	No Non-Rec. Investments + Reduced production cost + Increase concentration of Water in Lunar Regolith + Double demand + Price Increase	Add 1.25x Price

Table 4.3. Model results (key financial metrics) by version for Architectures 1 and 2.

	Year 1 Return on Equity		Project Rate of Return		Net Present Value	
	Arch 1	Arch 2	Arch 1	Arch 2	Arch 1	Arch 2
Version 0	N/A	N/A	N/A	N/A	\$ (5,275)	\$ (5,006)
Version 1	-30.3%	-30.5%	-11.9%	-11.9%	\$ (553)	\$ (561)
Version 2	-9.8%	-10.1%	-5.0%	-5.2%	\$ 255	\$ 240
Version 3	-2.3%	1.6%	-1.7%	-0.3%	\$ 593	\$ 726
Version 4	15.0%	15.2%	6.2%	5.9%	\$ 2,484	\$ 2,461
Version 5	26.1%	26.3%	12.8%	12.6%	\$ 4,156	\$ 4,134

Table 4.4. Financial statements for Version 5 of Architectures 1 and 2.

Architecture 1 - Financial Statements

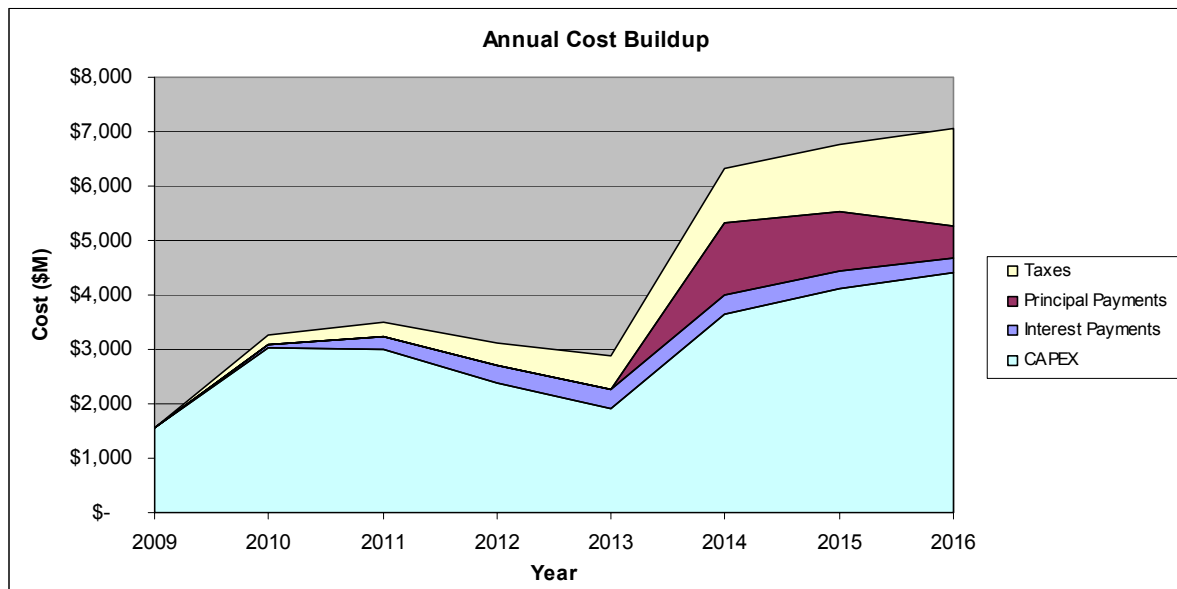
INCOME STATEMENT	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	Cumulative
Revenues	\$ 0	\$ 0	\$ 0	\$ 750	\$ 1,500	\$ 2,250	\$ 3,000	\$ 4,500	\$ 6,000	\$ 7,500	\$ 25,501
Gross Profit	\$ 0	\$ 0	\$ 0	\$ 689	\$ 1,378	\$ 2,067	\$ 2,755	\$ 4,133	\$ 5,511	\$ 6,888	\$ 23,421
EBITDA	\$ (4)	\$ (9)	\$ (10)	\$ 677	\$ 1,365	\$ 2,054	\$ 2,742	\$ 4,119	\$ 5,496	\$ 6,873	\$ 23,305
EBIT	\$ (4)	\$ (9)	\$ (10)	\$ 520	\$ 908	\$ 1,357	\$ 1,853	\$ 2,864	\$ 3,440	\$ 4,817	\$ 15,736
Net Income	\$ (4)	\$ (9)	\$ (10)	\$ 274	\$ 411	\$ 621	\$ 895	\$ 1,502	\$ 1,867	\$ 2,728	\$ 8,275
CASH FLOW	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	Cumulative
Net Cash From Operations	\$ (4)	\$ (9)	\$ (10)	\$ 431	\$ 868	\$ 1,317	\$ 1,783	\$ 2,758	\$ 3,924	\$ 4,784	\$ 15,844
Net Changes in Working Capital	\$ 0	\$ 0	\$ 0	\$ (57)	\$ (57)	\$ (57)	\$ (57)	\$ (115)	\$ (115)	\$ (115)	\$ (573)
CAPEX/NRE	\$ 0	\$ 0	\$ 1,587	\$ 2,998	\$ 2,993	\$ 2,394	\$ 1,923	\$ 3,670	\$ 4,127	\$ 3,880	\$ 23,571
Taxes	\$ -	\$ -	\$ -	\$ 167	\$ 274	\$ 414	\$ 596	\$ 1,002	\$ 1,245	\$ 1,819	\$ 5,517
Annual Cash (Shortfall) Surplus	\$ (4)	\$ (8)	\$ (1,596)	\$ (2,624)	\$ (2,182)	\$ (1,134)	\$ (197)	\$ (2,338)	\$ (1,409)	\$ 222	\$ (11,270)
Equity Financing	\$ 104	\$ 8	\$ 1,596	\$ 1,312	\$ 1,091	\$ 567	\$ 98	\$ 1,169	\$ 705	\$ -	\$ 6,650
Debt Financing	\$ -	\$ -	\$ -	\$ 1,312	\$ 1,091	\$ 567	\$ 98	\$ 1,169	\$ 705	\$ -	\$ 4,942
Principal and Interest Payments	\$ -	\$ -	\$ -	\$ 79	\$ 223	\$ 322	\$ 362	\$ 1,671	\$ 1,419	\$ 838	\$ 4,914
BALANCE SHEET	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	
Total Assets	\$ 100	\$ 100	\$ 1,686	\$ 4,589	\$ 7,187	\$ 8,947	\$ 10,043	\$ 12,582	\$ 14,778	\$ 16,950	
Short and Long Term Liabilities	\$ 0	\$ 1	\$ 1	\$ 1,318	\$ 2,414	\$ 2,986	\$ 3,089	\$ 2,957	\$ 2,581	\$ 2,024	
Shareholder Equity	\$ 104	\$ 112	\$ 1,708	\$ 3,020	\$ 4,111	\$ 4,678	\$ 4,776	\$ 5,945	\$ 6,650	\$ 6,650	
Retained Earnings	\$ (4)	\$ (13)	\$ (23)	\$ 251	\$ 662	\$ 1,283	\$ 2,178	\$ 3,680	\$ 5,547	\$ 8,275	

Architecture 2 - Financial Statements

INCOME STATEMENT	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	Cumulative
Revenues	\$ 0	\$ 0	\$ 0	\$ 750	\$ 1,500	\$ 2,250	\$ 3,000	\$ 4,500	\$ 6,000	\$ 7,500	\$ 25,501
Gross Profit	\$ 0	\$ 0	\$ 0	\$ 689	\$ 1,378	\$ 2,067	\$ 2,755	\$ 4,133	\$ 5,511	\$ 6,888	\$ 23,421
EBITDA	\$ (4)	\$ (9)	\$ (10)	\$ 677	\$ 1,365	\$ 2,054	\$ 2,742	\$ 4,119	\$ 5,496	\$ 6,873	\$ 23,305
EBIT	\$ (4)	\$ (9)	\$ (10)	\$ 523	\$ 910	\$ 1,360	\$ 1,857	\$ 2,870	\$ 3,395	\$ 4,772	\$ 15,665
Net Income	\$ (4)	\$ (9)	\$ (10)	\$ 276	\$ 411	\$ 621	\$ 896	\$ 1,505	\$ 1,841	\$ 2,698	\$ 8,225
CASH FLOW	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	Cumulative
Net Cash From Operations	\$ (4)	\$ (9)	\$ (10)	\$ 429	\$ 866	\$ 1,315	\$ 1,781	\$ 2,755	\$ 3,942	\$ 4,799	\$ 15,865
Net Changes in Working Capital	\$ 0	\$ 0	\$ 0	\$ (57)	\$ (57)	\$ (57)	\$ (57)	\$ (115)	\$ (115)	\$ (115)	\$ (573)
CAPEX/NRE	\$ 0	\$ 0	\$ 1,548	\$ 3,018	\$ 3,013	\$ 2,384	\$ 1,910	\$ 3,649	\$ 4,105	\$ 4,410	\$ 24,039
Taxes	\$ -	\$ -	\$ -	\$ 168	\$ 274	\$ 414	\$ 597	\$ 1,004	\$ 1,228	\$ 1,798	\$ 5,483
Annual Cash (Shortfall) Surplus	\$ (4)	\$ (8)	\$ (1,557)	\$ (2,646)	\$ (2,204)	\$ (1,127)	\$ (187)	\$ (2,332)	\$ (1,379)	\$ (290)	\$ (11,735)
Equity Financing	\$ 104	\$ 8	\$ 1,557	\$ 1,323	\$ 1,102	\$ 564	\$ 93	\$ 1,166	\$ 690	\$ 145	\$ 6,753
Debt Financing	\$ -	\$ -	\$ -	\$ 1,323	\$ 1,102	\$ 564	\$ 93	\$ 1,166	\$ 690	\$ 145	\$ 5,083
Principal and Interest Payments	\$ -	\$ -	\$ -	\$ 79	\$ 225	\$ 325	\$ 364	\$ 1,684	\$ 1,428	\$ 840	\$ 4,945
BALANCE SHEET	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	
Total Assets	\$ 100	\$ 100	\$ 1,648	\$ 4,575	\$ 7,195	\$ 8,949	\$ 10,037	\$ 12,561	\$ 14,690	\$ 17,124	
Short and Long Term Liabilities	\$ 0	\$ 1	\$ 1	\$ 1,329	\$ 2,437	\$ 3,005	\$ 3,104	\$ 2,957	\$ 2,555	\$ 2,146	
Shareholder Equity	\$ 104	\$ 112	\$ 1,670	\$ 2,993	\$ 4,095	\$ 4,659	\$ 4,752	\$ 5,918	\$ 6,608	\$ 6,753	
Retained Earnings	\$ (4)	\$ (13)	\$ (23)	\$ 253	\$ 664	\$ 1,285	\$ 2,181	\$ 3,686	\$ 5,528	\$ 8,225	

Note that the path of relaxed assumptions that was followed to improve the financial results of each successive version is not necessarily an optimal path. Numerous other variables could have been selected for relaxation, such as tax rates, hardware replacement rates, operations costs, discount rate, cost of debt and market capture rate. In addition, reduction factors could be applied to variables beyond production cost.

Figure 4.7. Cost buildup for Architecture 2, Version 5.



4.3.3 SENSITIVITY ANALYSIS

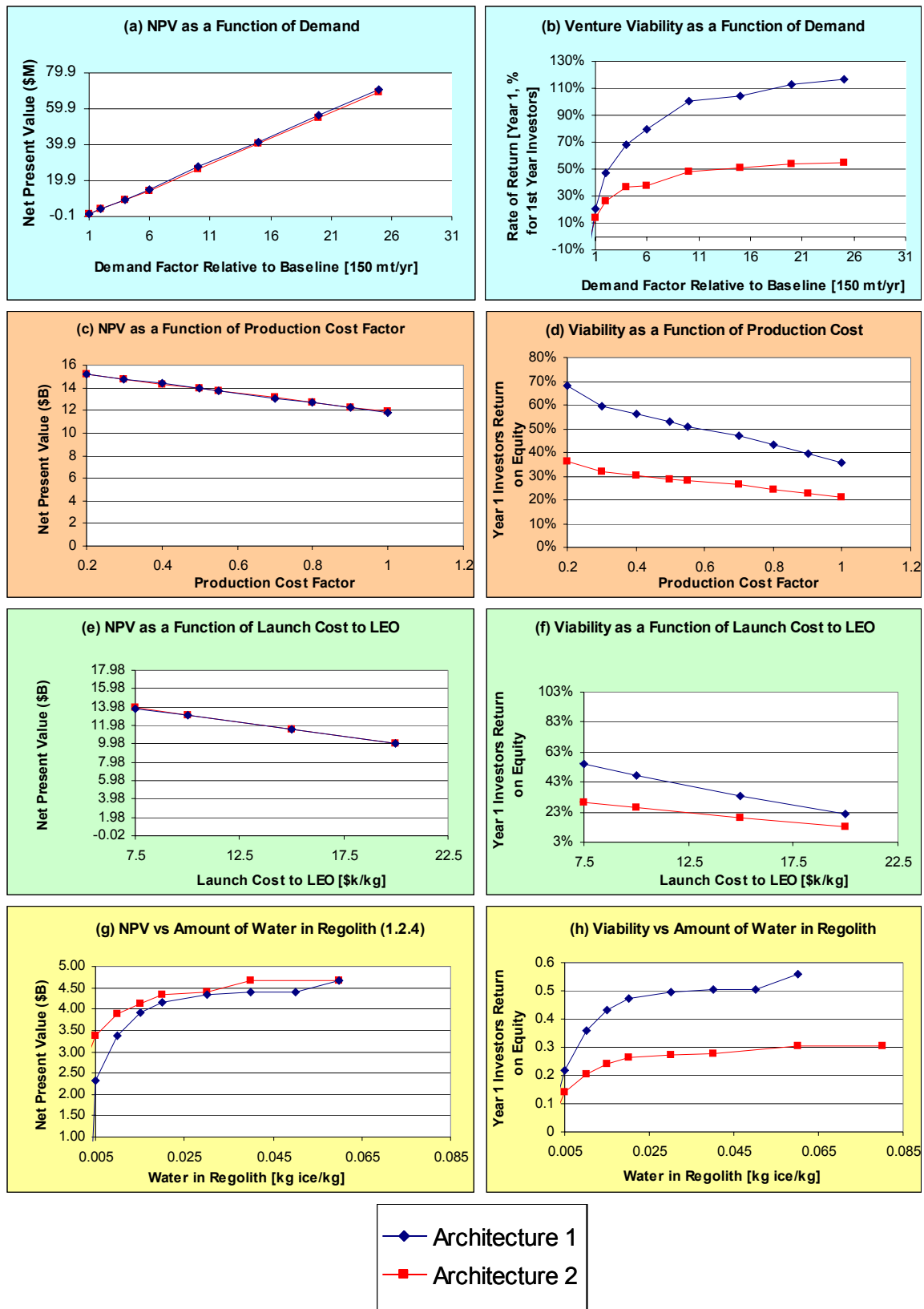
Sensitivity analysis was conducted on the Version 4 model, providing insight into the conditions under which the venture might be viewed as a good private sector investment. For example, the sensitivity to demand (Fig. 4.8a, b) shows that the venture would become viable for a fivefold increase in demand with respect to the baseline commercial LEO-to-GEO forecast. Other potential customers, such as military GEO satellites, solar system exploration missions by space agencies, and new markets such as orbital debris removal and/or avoidance, should be evaluated in future versions.

The sensitivity to production cost (Fig. 4.8c, d) can help identify target performance for technology development as well as production chain efficiency. In this case however, unrealistic improvements would be required to ensure financial viability. This can be interpreted in two ways: (1) Although they might help, production cost reduction efforts might not be the first priority for lunar resource development, or (2) A new scenario is required with much simpler design or break-through improvements in technology to enable a factor of five cost reduction.

The sensitivity to launch cost to LEO (Fig. 4.8e, f) shows how non-intuitive results can also be reached. “What if launch costs were much cheaper?” is a typical question when trying to improve the prospects for space business. However, the launch segment is not only a provider of service, but also a competitor. The net result is that financial viability actually decreases with decreasing launch cost to LEO.

Finally, Figs. 4.8g and h shows how the viability of the venture increases with water concentration in lunar regolith. This shows how the modeling approach can be used to provide a justification for exploration missions, and more generally the value of potential NASA’s actions to mitigate sources of uncertainty.

Figure 4.8. Sensitivity analysis of Version 5 of Architectures 1 and 2.



4.4 IMPLICATIONS FOR HUMAN EXPLORATION AND TECHNOLOGY

One metric that may be of interest to the human exploration community is the expected unit cost for fuel at the various production points. The unit costs presented in Figures 4.9 and 4.10 represent an *upper bound* on the expected cost of one unit of the resource at each destination (this is because the metric shown is total unit cost - fixed plus variable costs). Economic theory of natural resources predicts that a firm will continue to sell a product at a price just above the marginal costs (which can be approximated by variable costs alone), which are substantially lower than shown below. Note also that the unit costs below carry a heavy capital burden due to expanding capacity by increasing the size of the plant each year. Therefore, it is a reasonable expectation that a human exploration mission arriving at the L1 point could purchase fuel at a cost ranging between \$15 and \$5 Million per ton of fuel, depending on the ‘maturity’ (year of operation) of the commercial enterprise (note that the current expected cost is roughly \$35 Million per ton). Note that the marginal cost has not been calculated in the current model.

Figure 4.9. Unit costs at the Moon and L1 for Architecture 1, Version 5.

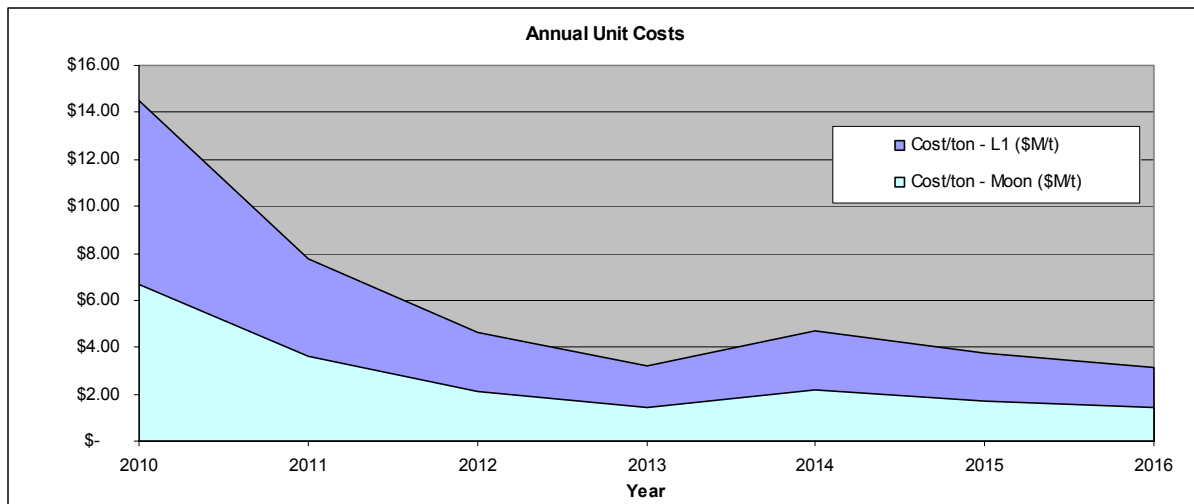
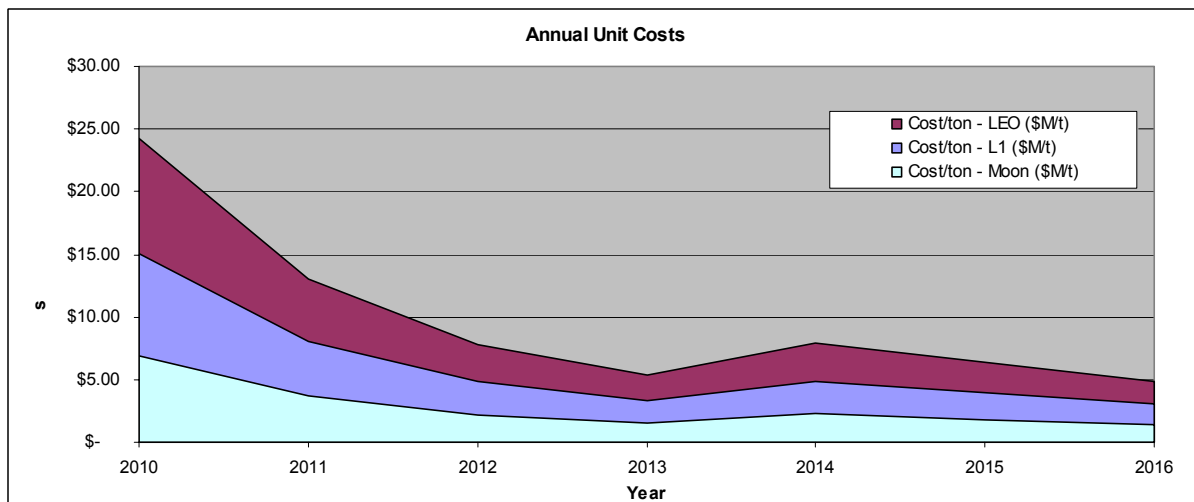


Figure 4.10. Unit costs at the Moon, L1 and LEO for Architecture 2, Version 5.



From a cursory inspection of the integrated financial model for each of the two architectures, several additional variables can be identified that have strong implications with respect to commercial viability, technology investment and human exploration:

1. The abundance of ice in the lunar regolith. The baseline case assumes that there is 1% ice in the regolith. If the ice concentration is higher in local areas, the amount of regolith that must be excavated and processed is proportionally less, and the amount of thermal energy required for its extraction is reduced. However, the power required for electrolysis and liquefaction remains the same because the total amount of water produced would remain constant. Figure 4.8(g) compares the net present value derived from the economic model for Architecture 2 Version 5. This comparison provides a prima facie case for the potential economic importance of further lunar exploration.
2. The mass of the lunar excavators and extractors. The designs for these elements that must work in the lunar shadowed craters are poorly understood. The current model may underestimate the effect of operating under extreme conditions (this could be corrected by modeling risk and reliability factors).
3. Power system architecture. The current baseline model assumes nuclear power systems, which currently are estimated to have specific masses of about 30 kg/kW. However, recent designs of thin film solar cell arrays have specific masses in the range of 1kg/kW. An architecture that utilizes solar energy could be reasonable for the polar application. Within relatively short distances of areas that apparently contain ice, high points with access to power most of the time exist (Bussey et al., 1999). Choices must be made as to the surface configuration of the power systems (they must be erected vertically because the sunlight is coming horizontally to the surface near the poles) and the means of transporting energy from sunlit areas into the shadow. If the specific mass of the total power system could be reduced to 5 kg/kW, significant reductions in transportation costs could result. In addition, NAFCOM costs nuclear systems at relatively high price/kg. Thin film solar arrays would have significantly lower costs if carried from Earth, and might even potentially be produced on the Moon.
4. The space transportation system. Masses for transportation elements have been derived from various literature sources, and are based on past designs. These may not reflect the best materials or technologies in the current application. The mass fraction of propellant that can be carried by a vehicle in space is quite sensitive to the dry mass of the spacecraft. If turning to new materials or technology can reduce structure, tanks, or other subsystem mass the effectiveness and profitability of a commercial architecture will improve.
5. Assumptions of system lifetime are also important for the space transportation elements. 10% per year refurbishment for all systems (e.g., mining plant, depot and spacecraft) has been assumed in this model, and each OTV is assumed to fly a mission once per month. This is approximately equivalent to an assumption that each vehicle can fly 120 times before being fully replaced (the validity of this assumption remains to be proven). Note that in Year 10 of

both models, refurbishment mass has risen to 1/3 of the total launch mass, and has become a significant cost factor.

6. Cost model assumptions. The NAFCOM cost model may have overestimated the development and production costs for the hardware, especially if a commercial development and procurement paradigm is assumed. Most of the analogies used in the current architectures are for single spacecraft or systems, principally built for government programs and therefore may not be applicable. This has been modeled by assuming that development and operations costs are some fraction of the NAFCOM costs (e. g. 70% - see integrated models Versions 2, 3, 4 and 5, for both architectures).
7. Cost optimization for the unit size of system elements in each architecture could minimize total costs by balancing up front (design and development) expenditures with long-run hardware costs. This kind of optimization was not conducted, and would require 'NAFCOM-like' parametric cost equations in optimization-friendly software (such as MS Excel). Due to the lack of transparency for NAFCOM cost engineering relationships, costs were estimated individually for each architectural variant in the current model.
8. The current model does not include any cost reduction associated with the possible commonality between elements used in different applications. For example, an OTV used for L1-LEO-L1 may have much in common with a lunar water tanker, in that the delta V requirements are similar and the systems might be very similar, with the exception of the landing gear on the lunar lander. Although differences will exist in the design of electrolysis units for lunar surface and 0-g applications, common development might lead to lower costs.
9. The current architecture was built to test a specific commercial market application. Additional markets, including government markets could be included and would raise the net present value of the operation. In particular, sales of propellant on the Moon or in L1 to Moon and Mars human exploration programs could provide significant benefits to those programs and an early human exploration program could choose to develop most of the systems identified in these architectures, allowing a later commercial opportunity to be developed at much lower cost.

5.0 CONCLUSIONS AND RECOMMENDATIONS FOR FUTURE WORK

We have developed and are improving an integrated engineering and financial modeling approach to enable rapid analysis of the financial viability of any space resource development venture. The approach consists in starting from a customer's point of view and a demand analysis, developing initial architectural concepts and modeling their scaling laws, and optimizing the scenario for the metrics of interest to private sector investors. We illustrated the advantages of this approach on a high-level lunar-propellant-based transportation service case study. "What if?" studies and sensitivity analysis help yield conclusions on the value of exploration missions and technology development, the optimal technical and business strategies, as well as the best public incentives to foster private sector involvement.

This modeling approach can be applied to other case studies, such as lunar mining for precious minerals, power production, solar cell production, and tourism; asteroid mining for water or precious minerals; in-space manufacturing for high-value materials or support of space endeavors; in-space transport using nuclear or solar electric propulsion; on-orbit servicing in Earth orbit and beyond; remote-sensing data commercialization; space tourism, and more. Application on such a wide space of possible ventures, and on different time scales can help draw a global map of the possible space resource development pathways for an integrated public and private sector space exploration strategy.